



Tutorial S1A: Permutations and Combinations

Section A (Basic Questions)

- 1 From a group of 5 women and 7 men, one of whom is Mr Lee, find how many committees of size 4 can be formed in which
- (i) there are 2 men and 2 women,
 - (ii) there is at least 1 man and Mr Lee is in the committee,
 - (iii) there is at least 1 man and at least 1 woman.

[(i) 210, (ii) 165, (iii) 455]

Solution:														
(i)	Number of committees with 2 men and 2 women = ${}^7C_2 \times {}^5C_2 = 210$													
(ii)	Since the committee must consist of at least one man and includes Mr Lee, we choose 3 people from the remaining 11 people. Number of ways = ${}^{11}C_3 = 165$													
(iii)	Listing Method: <table border="1"> <thead> <tr> <th>Women (5)</th><th>Men (7)</th><th>Number of ways</th></tr> </thead> <tbody> <tr> <td>1</td><td>3</td><td>${}^5C_1 \times {}^7C_3 = 175$</td></tr> <tr> <td>2</td><td>2</td><td>${}^5C_2 \times {}^7C_2 = 210$</td></tr> <tr> <td>3</td><td>1</td><td>${}^5C_3 \times {}^7C_1 = 70$</td></tr> </tbody> </table> Total number of ways is $175 + 210 + 70 = 455$	Women (5)	Men (7)	Number of ways	1	3	${}^5C_1 \times {}^7C_3 = 175$	2	2	${}^5C_2 \times {}^7C_2 = 210$	3	1	${}^5C_3 \times {}^7C_1 = 70$	We list all cases that give at least 1 man and at least 1 woman. To choose 4 people, we choose from women and from men separately. So, we use the multiplication principle: (Number of ways to choose 4 people) = (Number of ways to choose from women) $\times$ (Number of ways to choose from men) Any selection of at least 1 man and at least 1 woman must fall into either Case 1 or Case 2 or Case 3. Since these cases are mutually exclusive, we apply the addition principle.
Women (5)	Men (7)	Number of ways												
1	3	${}^5C_1 \times {}^7C_3 = 175$												
2	2	${}^5C_2 \times {}^7C_2 = 210$												
3	1	${}^5C_3 \times {}^7C_1 = 70$												
	Complement Method: Number of committees with at least 1 man and at least 1 woman = (Number of committees without restriction) – (Number of all men committees) – (Number of all women committees) $= {}^{12}C_4 - {}^7C_4 - {}^5C_4 = 455$	Similar to Example 11(ii) in the notes. Note: ${}^5C_1 \times {}^7C_1 \times {}^{10}C_2$ is <u>incorrect</u>; there is double counting. For example, $W_1 M_1 W_2 M_2$ and $W_2 M_1 W_1 M_2$ are the same selection but counted as 2 different ways if we use the above wrong method.												

2 9 coloured balls are arranged in a row. Find the number of ways this can be done if

- (i) all the balls are of different colours,
- (ii) there are 4 red balls, 3 green balls and 2 yellow balls,
- (iii) there are 4 red balls, 3 green balls and 2 yellow balls, **and** the arrangement is symmetrical.

[(i) 362880, (ii) 1260, (iii) 12]

Solution:

(i) Number of ways = $9! = 362880$

(ii) 9 coloured balls: 4 R, 3 G, 2 Y
 Number of ways = $\frac{9!}{4!3!2!} = 1260$

Alternative method:

First choose 4 out of the 9 positions for the red balls. Then of the remaining 5 positions, choose 3 for the green balls. The last two positions must go to the yellow balls. Hence number of ways is ${}^9C_4 \times {}^5C_3$

(iii)

_____ $\frac{G}{2R, 1G, 1Y}$ _____ $\frac{G}{2R, 1G, 1Y}$ _____

Since arrangement is symmetrical, we first consider the middle ball, then arrange the first 4 balls, and the arrangement of the last 4 balls will be determined by that of the first 4 balls.

The middle ball must be green, so there's only 1 way for the middle ball.

The first 4 balls consist of 2 red, 1 green and 1 yellow ball. The number of ways to arrange these 4 balls is $\frac{4!}{2!}$. (Or: First choose 2 positions for the red balls. Then choose 1 position for

the green ball. So, number of ways to arrange the 4 balls is ${}^4C_2 \times {}^2C_1$)

Since arrangement is symmetrical, once we arrange the first 4 balls, there's only 1 way to arrange the last 4 balls.

Hence required number of ways is $1 \times \frac{4!}{2!} \times 1 = 12$

Number of ways
for middle ball

Number of ways
for 1st 4 balls

Number of ways
for last 4 balls

- 3 How many different three figure numbers between 100 and 999 inclusive, contain two and only two consecutive identical figures?

[162]

Solution:

Since the number must contain only two consecutive identical figures, either the first 2 digits are identical, or the last 2 digits are identical.

Case 1: First 2 digits are identical.

The first 2 digits can be 11, 22, 33, ..., 99, so there are 9 ways for the 1st 2 digits.

The 3rd digit can be any digit from 0 to 9 that has not been used for 1st digit, so there are 9 ways for the 3rd digit.

Number of ways = $9 \times 9 = 81$

Case 2: Last 2 digits are identical

The 1st digit can be 1, 2, 3, ..., 9, so there are 9 ways for the 1st digit

The last 2 digits can be any digit from 0 to 9 that has not been used for 1st digit, so there are 9 ways for the last 2 digits.

Number of ways = $9 \times 9 = 81$

Total number of three figure numbers = $81 + 81 = 162$

- 4 (a) A resort hotel has 3 single rooms, 5 double rooms and 4 family rooms. On a particular week, 2 individuals book a single room each, 3 couples book a double room each and 3 families book a family room each. Given that all the rooms are available for that week and each room has a distinct design theme, find the number of different possible bookings of rooms amongst the guests.
- (b) A family of 6 adults and 3 children are to be seated at a round table. Find the number of possible arrangements such that
- all the 3 children sit together,
 - none of the children sit together.

[(a) 8640; (b)(i) 4320, (ii) 14400]

NJC Prelim 9233/2005/01/Q7 (modified)

Solution:

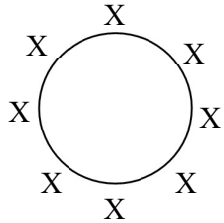
(a)	Rooms: 3 single, 5 double, 4 family People: 2 individual, 3 couples, 3 families For the individuals, there are 3 single rooms available that can be assigned to the 2 individuals, hence the number of possible bookings for the individuals is 3P_2 or ${}^3C_2 \times 2!$ Similarly, the number of possible bookings for the couples is 5P_3 or ${}^5C_3 \times 3!$ And the number of possible bookings for families is 4P_3 or ${}^4C_3 \times 3!$ Total number of possible bookings is ${}^3P_2 \times {}^5P_3 \times {}^4P_3 = 8640$	We need to find the number of possible bookings for the individuals and the number of possible bookings for the couples and the number of possible bookings for the families. Hence the final answer is obtained by multiplying the number of ways for each category.
(b)(i)	All 3 children sit together, so consider them as 1 unit. There are 6 adults, so altogether we have 7 units to be arranged around a table. $\textcircled{C C C} A A A A A A - 7 \text{ units}$ Number of ways to arrange 7 units at a round table = $(7 - 1)!$ Number of arrangements within the “children unit” = $3!$ Total number of seating arrangements = $(7 - 1)! \times 3! = 4320$	
(b)(ii)	To separate the children, we first arrange the adults, then slot/arrange the children in between the adults. So, we use the <u>Slotting Method</u>: Number of arrangements to first seat the 6 adults = $(6 - 1)! = 120$ Number of arrangements to slot in the 3 children in the 6 empty spaces between the adults = 6P_3 or ${}^6C_3 \times 3!$ Total number of seating arrangements = $(6 - 1)! \times {}^6P_3 = 14400$	

- 5 Six people sit at a round table with eight identical chairs. Find the number of seating arrangements if
- (i) there are no restrictions,
 - (ii) the seats are numbered.

[(i) 2520, (ii) 20160]

TJC Prelim 9740/2011/02/Q6(b)

Solution:

(i)	Since there are 6 people and 8 chairs, 2 chairs would not be occupied. Number of ways to arrange 6 people and 2 empty slots (if the 2 slots are distinct) at a round table = $(8 - 1)! = 7!$ <i>However, the 2 empty slots are actually identical.</i> Hence, number of arrangements = $\frac{7!}{2!} = 2520$	
(ii)	If the seats are numbered, number of arrangements = $2520 \times 8 = 20160$	If seats are numbered, we first consider the number of arrangements if seats are identical, then multiply by the number of seats: (Number of arrangements if seats <u>are numbered</u>) = (Number of arrangements if seats <u>are not numbered</u>) $\times$ (Number of seats)

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